

TUOYI (toripalimab) 240mg/6ml (40mg/ml) Concentrate for Solution for Infusion

1. NAME OF THE MEDICINE PRODUCT

TUOYI (toripalimab) 240mg/6ml (40mg/ml) Concentrate for Solution for Infusion

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Toripalimab is a programmed death receptor-1 (PD-1) blocking antibody. Toripalimab is a humanized immunoglobulin G4 (IgG4) kappa immunoglobulin that has a predicted molecular weight of approximately 150 kDa. It is expressed in a recombinant Chinese Hamster Ovary (CHO) mammalian cell expression system.

TUOYITM (toripalimab) injection is a sterile, preservative-free, clear to slightly opalescent, colorless to slightly yellow solution for intravenous use supplied in a single-dose vial. Each vial contains 240 mg of TUOYITM in 6 mL of solution. Each mL of solution contains 40 mg toripalimab, citric acid monohydrate (0.51 mg), mannitol (25 mg), polysorbate 80 (0.20 mg), sodium chloride (2.92 mg), sodium citrate (4.65 mg), and Water for Injection, USP, at pH 6.0.

3. PHARMACEUTICAL FORM

Injection: 240 mg/6 mL (40 mg/mL) clear to slightly opalescent, colourless to slightly yellow solution in a single-dose vial.

4. CLINICAL PARTICULARS

4.1. Therapeutic indications

First-line Treatment of Metastatic or Recurrent, Locally Advanced NPC with Cisplatin and Gemcitabine

TUOYITM is indicated, in combination with cisplatin and gemcitabine, for the first-line treatment of adults with metastatic or with recurrent, locally advanced nasopharyngeal carcinoma (NPC).

Previously Treated Unresectable or Metastatic NPC

TUOYITM is indicated, as a single agent, for the treatment of adults with recurrent unresectable or metastatic NPC with disease progression on or after a platinum-containing chemotherapy.

4.2. Posology and method of administration

Recommended Dosage

The recommended dosages of TUOYITM are provided in Table 1. Administer as recommended.

Table 1: Recommended Dosage

Indication	Recommended Dosage of TUOYI™	Duration of Treatment
First-line NPC	240 mg every three weeks	Until disease progression, unacceptable toxicity, or up to 24 months.
Recurrent NPC	3 mg/kg every two weeks	Until disease progression or unacceptable toxicity.

Dosage Modifications

No dose reductions of TUOYI™ are recommended. In general, withhold TUOYI™ for severe (Grade 3) immune-mediated adverse reactions. Permanently discontinue TUOYI™ for life-threatening (Grade 4) immune-mediated adverse reactions, recurrent severe (Grade 3) immune-mediated reactions that require systemic immunosuppressive treatment, or an inability to reduce prednisone to 10 mg per day or less (or equivalent) within 12 weeks of initiating steroids.

Dosage modifications for TUOYI™ for adverse reactions that require management different from these general guidelines are summarized in Table 2.

Table 2: Recommended Dosage Modifications for Adverse Reactions

Adverse Reaction	Severity ¹	Dose Modification
Immune-Related Adverse Reactions		
Pneumonitis	Grade 2	Withhold ²
	Grades 3 or 4	Permanently discontinue
Colitis	Grade 2 or 3	Withhold ²
	Grade 4	Permanently discontinue
Hepatitis with no tumor involvement of the liver	AST/ALT increases to more than 3 and up to 8 times ULN or Total bilirubin increases to more than 1.5 and up to 3 times ULN	Withhold ²
	AST or ALT increases to more than 8 times ULN or Total bilirubin increases to more than 3 times ULN	Permanently discontinue
Hepatitis with tumor involvement of the liver ³	Baseline AST or ALT is more than 1 and up to 3 times ULN and increases to more than 5 and up to 10 times ULN or Baseline AST or ALT is more than 3 and up to 5 times ULN and increases to more than 8 and up to 10 times ULN	Withhold ²

Adverse Reaction	Severity ¹	Dose Modification
Immune-Related Adverse Reactions		
	Baseline AST or ALT is above the ULN and increases to more than 10 times ULN or Total bilirubin increases to more than 3 times ULN	Permanently discontinue
Endocrinopathies	Grades 3 or 4	Withhold until clinically stable or permanently discontinue depending on severity ²
Nephritis with Renal Dysfunction	Grade 2 or 3 increased blood creatinine	Withhold ²
	Grade 4 increased blood creatinine	Permanently discontinue
Exfoliative Dermatologic Conditions	Suspected SJS, TEN, or DRESS	Withhold ²
	Confirmed SJS, TEN, or DRESS	Permanently discontinue
Myocarditis	Grades 2, 3, or 4	Permanently discontinue
Neurological toxicities	Grade 2	Withhold ²
	Grade 3-4	Permanently discontinue
Other Adverse Reactions		
Infusion-related reactions	Grade 1 or 2	Interrupt or slow the rate of infusion
	Grade 3 or 4	Stop infusion Permanently discontinue

¹ Based on National Cancer Institute (NCI) Common Terminology for Adverse Events (CTCAE) version 5.0

² Resume TUOYI™ in patients with complete or partial resolution to Grade 0-1 after corticosteroid taper. Permanently discontinue if no complete or partial resolution within 12 weeks of initiating steroids or inability to reduce prednisone to 10 mg per day or less (or equivalent) within 12 weeks of initiating steroids.

³ If AST and ALT are less than or equal to ULN at baseline in patients with liver involvement, withhold or permanently discontinue TUOYI™ based on recommendations for hepatitis with no liver involvement.

ALT=alanine aminotransferase, AST=aspartate aminotransferase, DRESS=drug rash with eosinophilia and systemic symptoms, SJS=Stevens Johnson syndrome, TEN=toxic epidermal necrolysis, ULN=upper limit of normal.

Preparation and Administration

Preparation for Intravenous Infusion

- Visually inspect the solution for particulate matter and discoloration. The solution is clear to slightly opalescent, colorless to slightly yellow. Discard the vial if visible particles are observed.
- Withdraw the required volume of TUOYI™ and inject slowly into a 100 mL or 250 mL infusion bag containing 0.9% Sodium Chloride Injection, USP. **Mix diluted solution by gentle inversion.** Do not shake. The final concentration of the diluted solution should be between 1 mg/mL to 3 mg/mL.
- Discard any unused portion left in the vial.

Storage of Diluted Solution for Infusion

TUOYI™ does not contain a preservative.

If the diluted solution is not administered immediately, store either:

- At room temperature, 20°C to 25°C, for no more than 8 hours from the time of dilution to the completion of the infusion. Discard diluted solution stored at room temperature after 8 hours. Or
- Refrigerated at 2°C to 8°C, for no more than 24 hours from the time of dilution to the completion of the infusion. If refrigerated, allow the diluted solution to come to room temperature prior to administration. Discard the refrigerated diluted solution after 24 hours.

Do not freeze.

Administration

- Administer diluted solution intravenously via an infusion pump using an in-line aseptic filter (0.2 or 0.22 micron).
- First Infusion: Infuse over at least 60 minutes.
- Subsequent infusions: If no infusion-related reactions occurred during the first infusion, subsequent infusions may be administered over 30 minutes.
- Do not co-administer other drugs through the same intravenous line.
- When administered on the same day as chemotherapy, TUOYI™ should be administered prior to chemotherapy.
- Refer to the Prescribing Information for cisplatin and gemcitabine for recommended dosing information.

4.3. Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

4.4. Special warnings and precautions for use

Traceability

In order to improve the traceability of biological medicinal products, the name and the batch number of the administered product should be clearly recorded.

Severe and Fatal Immune-Mediated Adverse Reactions

TUOYI™ is a monoclonal antibody that belongs to a class of drugs that bind to either the programmed death receptor-1 (PD-1) or PD-ligand 1 (PD-L1), blocking the PD-1/PD-L1 pathway, thereby removing inhibition of the immune response, potentially breaking peripheral tolerance and inducing immune-mediated adverse reactions. Important immune-mediated

adverse reactions listed under WARNINGS AND PRECAUTIONS may not include all possible severe and fatal immune-mediated reactions.

Immune-mediated adverse reactions, which may be severe or fatal, can occur in any organ system or tissue and can affect more than one body system simultaneously. Immune-mediated adverse reactions can occur at any time after starting PD-1/PD-L1 blocking antibody. While immune-mediated adverse reactions usually manifest during treatment with PD-1/PD-L1 blocking antibodies, immune-mediated adverse reactions can also manifest after discontinuation of PD-1/PD-L1 blocking antibodies.

Early identification and management of immune-mediated adverse reactions are essential to ensure safe use of PD-1/PD-L1 blocking antibodies. Monitor closely for symptoms and signs that may be clinical manifestations of underlying immune-mediated adverse reactions. Evaluate liver enzymes, creatinine, and thyroid function at baseline and periodically during treatment. In cases of suspected immune-mediated adverse reactions, initiate appropriate workup to exclude alternative etiologies, including infection. Institute medical management promptly, including specialty consultation as appropriate.

Withhold or permanently discontinue TUOYI™ depending on severity. In general, if TUOYI™ requires interruption or discontinuation, administer systemic corticosteroid therapy (1 to 2 mg/kg/day prednisone or equivalent) until improvement to Grade 1 or less. Upon improvement to Grade 1 or less, initiate corticosteroid taper and continue to taper over at least 1 month. Consider administration of other systemic immunosuppressants in patients whose immune-mediated adverse reactions are not controlled with corticosteroid therapy.

Toxicity management guidelines for adverse reactions that do not necessarily require systemic steroids (e.g., endocrinopathies and dermatologic reactions) are discussed below.

Immune-Mediated Pneumonitis

TUOYI™ in Combination with Cisplatin and Gemcitabine

TUOYI™ in combination with cisplatin and gemcitabine can cause immune-mediated pneumonitis. In patients treated with other PD-1/PD-L1 blocking antibodies, the incidence of pneumonitis is higher in patients who have received prior thoracic radiation. Immune-mediated pneumonitis occurred in 2.1% (3/146) of patients receiving TUOYI™, including Grade 2 (1.4%) adverse reactions. Pneumonitis resolved in 67% (2/3) of these patients.

TUOYI™ as a Single-Agent

TUOYI™ can cause immune-mediated pneumonitis. In patients treated with other PD-1/PD-L1 blocking antibodies, the incidence of pneumonitis is higher in patients who have received prior thoracic radiation. Immune-mediated pneumonitis occurred in 2.6% (22/851) of patients receiving TUOYI™, including fatal (0.2%), Grade 3 (0.7%), and Grade 2 (1.1%) adverse reactions. Systemic corticosteroids were required in 82% (18/22) of patients with pneumonitis. Pneumonitis led to permanent discontinuation of TUOYI™ in 1.2% (10/851) of patients. Pneumonitis resolved in 23% (5/22) of these patients.

Immune-Mediated Colitis

TUOYI™ can cause immune-mediated colitis, which may present with diarrhea. Cytomegalovirus (CMV) infection/reactivation has been reported in patients with corticosteroid-refractory immune-mediated colitis. In cases of corticosteroid-refractory colitis, consider repeating infectious workup to exclude alternative etiologies.

TUOYI™ as a Single-Agent

Immune-mediated colitis occurred in 0.4% (3/851) of patients receiving TUOYI™, including Grade 3 (0.2%) and Grade 2 (0.1%) adverse reactions. Colitis resolved in all 3 patients.

Hepatotoxicity and Immune-Mediated Hepatitis

TUOYI™ in Combination with Cisplatin and Gemcitabine

TUOYI™ in combination with cisplatin and gemcitabine can cause immune-mediated hepatitis. Immune-mediated hepatitis occurred in 0.7% (1/146) of patients receiving TUOYI™ in combination with cisplatin and gemcitabine, which was a Grade 3 (0.7%) adverse reaction. The patient with immune-mediated hepatitis required systemic corticosteroids.

TUOYI™ as a Single-Agent

TUOYI™ can cause immune-mediated hepatitis. Immune-mediated hepatitis occurred in 3.3% (28/851) of patients receiving TUOYI™, including Grade 4 (0.8%), Grade 3 (2.1%), and Grade 2 (0.4%) adverse reactions. Hepatitis led to permanent discontinuation of TUOYI™ in 1.1% of patients and withholding of TUOYI™ in 0.8% of patients. Hepatitis resolved in 54% (15/28) of these patients.

Immune-Mediated Endocrinopathies

Adrenal Insufficiency

TUOYI™ can cause primary or secondary adrenal insufficiency. For Grade 2 or higher adrenal insufficiency, initiate symptomatic treatment, including hormone replacement as clinically indicated. Withhold or permanently discontinue TUOYI™ depending on severity.

TUOYI™ as a Single-Agent

Adrenal insufficiency occurred in 0.5% (4/851) of the patients receiving TUOYI™, including Grade 2 (0.4%) and Grade 1 (0.1%) adverse reactions. Systemic corticosteroids were required in 75% (3/4) of the patients with adrenal insufficiency. Adrenal insufficiency led to withholding of TUOYI™ in 0.1% (1/851) of patients. In the one patient in whom TUOYI™ was withheld, TUOYI™ was reinitiated after symptom improvement.

Hypophysitis

TUOYI™ can cause immune-mediated hypophysitis. Hypophysitis can present with acute symptoms associated with mass effects such as headache, photophobia, or visual field defects. Hypophysitis can cause hypopituitarism. Initiate hormone replacement as indicated. Withhold or permanently discontinue TUOYI™ depending on severity.

TUOYI™ as a Single-Agent

Hypophysitis occurred in 0.4% (3/851) of patients receiving TUOYI™, including Grade 3 (0.2%) and Grade 2 (0.1%) adverse reactions. All three patients received systemic corticosteroids. Hypophysitis led to permanent discontinuation of TUOYI™ in 0.1% (1/851) of patients and withholding of TUOYI™ in 0.1% (1/851) of patients. The one patient in whom TUOYI™ was withheld reinitiated TUOYI™.

Thyroid Disorders

TUOYI™ can cause immune-mediated thyroid disorders. Thyroiditis can present with or without endocrinopathy. Hypothyroidism can follow hyperthyroidism. Initiate hormone replacement for hypothyroidism or institute medical management of hyperthyroidism as clinically indicated. Withhold or permanently discontinue TUOYI™ depending on severity.

TUOYI™ in Combination with Cisplatin and Gemcitabine

Thyroiditis occurred in 2.1% (3/146) of patients receiving TUOYI™ in combination with cisplatin and gemcitabine, including Grade 2 (1.4%). Three patients required thyroid hormone replacement therapy. Thyroiditis resolved in one of the 3 patients.

Hyperthyroidism occurred in 1.4% (2/146) of patients receiving TUOYI™ in combination with cisplatin and gemcitabine. Hyperthyroidism resolved in these 2 patients.

Hypothyroidism occurred in 30% (44/146) of patients receiving TUOYI™ in combination with cisplatin and gemcitabine, including Grade 2 (24%) and Grade 1 (6%). Eighty percent of the 44 patients required thyroid hormone replacement therapy. TUOYI™ was withheld in 2.1% (3/146) of the patients. Of the 3 patients in whom TUOYI™ was withheld, 2 patients reinitiated TUOYI™.

TUOYI™ as a Single-Agent

Thyroiditis occurred in 0.6% (5/851) of patients receiving TUOYI™, including Grade 2 (0.1%). Two of these 5 patients received systemic corticosteroids and 2 required thyroid hormone replacement therapy. Thyroiditis resolved in 2 of the 5 patients.

Hyperthyroidism occurred in 7% (55/851) of patients receiving TUOYI™, including Grade 2 (1.9%). Hyperthyroidism resolved in 85% (47/55) of the patients.

Hypothyroidism occurred in 15% (128/851) of patients receiving TUOYI™, including Grade 2 (8%). Sixty three percent of the 128 patients required thyroid hormone replacement therapy. TUOYI™ was withheld in 0.5% of patients. Of the 4 patients in whom TUOYI™ was withheld, 3 patients reinitiated TUOYI™.

Type 1 Diabetes Mellitus, which can present with Diabetic Ketoacidosis

Monitor patients for hyperglycemia or other signs and symptoms of diabetes. Initiate treatment with insulin as clinically indicated. Withhold or permanently discontinue TUOYI™ depending on severity.

TUOYI™ as a Single-Agent

Diabetes mellitus occurred in 0.9% (8/851) of patients receiving TUOYI™, including Grade 4 (0.1%), Grade 3 (0.7%), and Grade 2 (0.1%). Diabetes mellitus led to permanent discontinuation in 0.4% of patients. Six of the 8 (75%) patients with diabetes mellitus required long-term insulin therapy.

Immune-Mediated Nephritis with Renal Dysfunction

TUOYI™ in Combination with Cisplatin and Gemcitabine

TUOYI™ in combination with cisplatin and gemcitabine can cause immune-mediated nephritis. Immune-mediated nephritis occurred in 0.7% (1/146) of patients receiving TUOYI™. The one patient with immune-mediated nephritis (Grade 4) required systemic corticosteroids and nephritis led to discontinuation of TUOYI™. Nephritis resolved in this patient.

TUOYI™ as a Single-Agent

TUOYI™ can cause immune-mediated nephritis. Immune-mediated nephritis occurred in 0.5% (4/851) of patients receiving TUOYI™, including Grade 3 (0.5%) adverse reactions. Nephritis resolved in 75% (3/4) of these patients.

Immune-Mediated Dermatologic Adverse Reactions

TUOYI™ can cause immune-mediated rash or dermatitis. Exfoliative dermatitis, including Stevens-Johnson Syndrome (SJS), drug rash with eosinophilia and systemic symptoms (DRESS), and toxic epidermal necrolysis (TEN), has occurred with PD-1/PD-L1 blocking antibodies. Topical emollients and/or topical corticosteroids may be adequate to treat mild to

moderate nonexfoliative rashes. Withhold or permanently discontinue TUOYI™ depending on severity.

TUOYI™ in Combination with Cisplatin and Gemcitabine

Immune-mediated dermatologic adverse reactions occurred in 8% (12/146) of patients receiving TUOYI™, including Grade 3 (3.4%) and Grade 2 (1.4%) adverse reactions. Systemic corticosteroids were required in 25% (3/12) of the patients with immune-mediated dermatologic adverse reactions. Immune-mediated dermatologic adverse reactions led to permanent discontinuation of TUOYI™ in 2.1% (3) of patients. Immune-mediated dermatologic adverse reactions resolved in 92% (11/12) of these patients.

TUOYI™ as a Single-Agent

Immune-mediated dermatologic adverse reactions occurred in 4% (34/851) of patients receiving TUOYI™, including Grade 3 (0.4%) and Grade 2 (1.4%) adverse reactions. Immune-mediated dermatologic adverse reactions led to withholding of TUOYI™ in 0.4% (3) of the patients. Systemic corticosteroids were required in 12% (4/34) of the patients with immune-mediated dermatologic adverse reactions. Immune-mediated dermatologic adverse reactions resolved in 71% (24/34) of these patients.

Other Immune-Mediated Adverse Reactions

The following clinically significant immune-mediated adverse reactions occurred at an incidence of <1% (unless otherwise noted) in patients who received TUOYI™ or were reported with the use of other PD1/PD-L1 blocking antibodies. Severe or fatal cases have been reported for some of these adverse reactions.

Cardiac/Vascular: Myocarditis, pericarditis, vasculitis, pericardial effusion

Nervous System: Meningitis, encephalitis, myelitis and demyelination, myasthenic syndrome/myasthenia gravis (including exacerbation), Guillain-Barré syndrome, nerve paresis, autoimmune neuropathy

Ocular: Uveitis, iritis and other ocular inflammatory toxicities can occur. Some cases can be associated with retinal detachment. Various grades of visual impairment, including blindness, can occur. If uveitis occurs in combination with other immune-mediated adverse reactions, consider a Vogt-Koyanagi-Harada-like syndrome, as this may require treatment with systemic steroids to reduce the risk of permanent vision loss.

Gastrointestinal: Pancreatitis, to include increases in serum amylase and lipase levels, gastritis, duodenitis

Musculoskeletal and Connective Tissue: Myositis/polymyositis, rhabdomyolysis (and associated sequelae, including renal failure), arthritis, polymyalgia rheumatica, dermatomyositis

Endocrine: Hypoparathyroidism

Hematologic/Immune: Hemolytic anemia, aplastic anemia, hemophagocytic lymphohistiocytosis, systemic inflammatory response syndrome, histiocytic necrotizing lymphadenitis (Kikuchi lymphadenitis), sarcoidosis, immune thrombocytopenic purpura, solid organ transplant rejection.

Infusion-Related Reactions

TUOYI™ can cause severe or life-threatening infusion-related reactions including hypersensitivity and anaphylaxis.

TUOYI™ in Combination with Cisplatin and Gemcitabine

Infusion-related reactions have been reported in 4.1% of patients receiving TUOYI™ in combination with cisplatin and gemcitabine, including Grade 2 (0.7%) reactions.

TUOYI™ as a Single-Agent

Infusion-related reactions occurred in 2% of 851 patients receiving TUOYI™ as single agent, including Grade 3 (0.1%) and Grade 2 (0.6%). TUOYI™ was withheld for one Grade 3 infusion related reaction.

Monitor patients for signs and symptoms of infusion-related reactions including rigors, chills, wheezing, pruritus, flushing, rash, hypotension, hypoxemia, and fever. Interrupt or slow the rate of infusion for mild (Grade 1) or moderate (Grade 2) infusion-related reactions. For severe (Grade 3) or life-threatening (Grade 4) infusion-related reactions, stop infusion and permanently discontinue TUOYI™.

Complications of Allogeneic HSCT

Fatal and other serious complications can occur in patients who receive allogeneic hematopoietic stem cell transplantation (HSCT) before or after being treated with a PD-1/PD-L1 blocking antibody. Transplant-related complications include hyperacute graft-versus-host-disease (GVHD), acute GVHD, chronic GVHD, hepatic veno-occlusive disease (VOD) after reduced intensity conditioning, and steroid-requiring febrile syndrome (without an identified infectious cause). These complications may occur despite intervening therapy between PD-1/PD-L1 blockade and allogeneic HSCT.

Follow patients closely for evidence of transplant-related complications and intervene promptly. Consider the benefit versus risks of treatment with a PD-1/PD-L1 blocking antibody prior to or after an allogeneic HSCT.

Embryo-Fetal Toxicity

Based on its mechanism of action, TUOYI™ can cause fetal harm when administered to a pregnant woman. Animal studies have demonstrated that inhibition of the PD-1/PD-L1 pathway can lead to increased risk of immune-mediated rejection of the developing fetus resulting in fetal death. Advise women of the potential risk to a fetus. Advise females of reproductive potential to use effective contraception during treatment with TUOYI™ and for 4 months after the last dose.

4.5. Interaction with other medicinal products and other forms of interaction

No formal interaction studies have been performed. Since toripalimab is cleared from the circulation through catabolism, no metabolic drug-drug interactions are expected. Toripalimab is not a substrate for cytochrome P450 or active substance transporters. Toripalimab is not a cytokine and is unlikely to be a cytokine modulator. Additionally, pharmacokinetic (PK) interaction of toripalimab with small molecule active substances is not expected. There is no evidence of interaction mediated by non-specific clearance of lysosome degradation for antibodies.

The use of systemic corticosteroids or immunosuppressants before starting toripalimab should be avoided because of their potential interference with the pharmacodynamic activity and efficacy of toripalimab. However, systemic corticosteroids or other immunosuppressants can be used after starting toripalimab to treat immune-related adverse reactions (see section 4.4). Corticosteroids can also be used as premedication, when toripalimab is used in combination with chemotherapy, as antiemetic prophylaxis and/or to alleviate chemotherapy-related adverse reactions.

4.6. Fertility, pregnancy and lactation

Pregnancy

Risk Summary

Based on its mechanism of action, TUOYI™ can cause fetal harm when administered to a pregnant woman [see Clinical Pharmacology (12.1)]. There are no available data on the use of TUOYI™ in pregnant women. Animal studies have demonstrated that inhibition of the PD1/PD-L1 pathway can lead to increased risk of immune-mediated rejection of the developing fetus and result in fetal death (see Data). Human IgG4 immunoglobulins (IgG4) are known to cross the placenta; therefore, TUOYI™ can potentially be transmitted from the mother to the developing fetus. Advise women of the potential risk to a fetus.

Data

Animal Data

Animal reproduction studies have not been conducted with TUOYI™ to evaluate its effect on reproduction and fetal development. A central function of the PD-1/PD-L1 pathway is to preserve pregnancy by maintaining maternal immune tolerance to the fetus. In murine models of pregnancy, blockade of PD-L1 signaling has been shown to disrupt tolerance to the fetus and to result in an increase in fetal loss; therefore, potential risks of administering TUOYI™ during pregnancy could include increased rates of abortion or stillbirth. As reported in the literature, there were no malformations related to the blockade of PD-1/PD-L1 signaling in the offspring of these animals; however, immune-mediated disorders occurred in PD-1 and PD-L1 knockout mice. Based on its mechanism of action, fetal exposure to toripalimab may increase the risk of developing immune-mediated disorders or altering the normal immune response.

Lactation

Risk Summary

There are no data on the presence of toripalimab in human milk or its effects on the breastfed child or on milk production. Maternal IgG is known to be present in human milk. The effects of local gastrointestinal exposure and limited systemic exposure in the breastfed child to toripalimab are unknown. Because of the potential for serious adverse reactions in breastfed children, advise lactating women not to breastfeed during treatment with TUOYI™ and for 4 months after the last dose.

Females and Males of Reproductive Potential

TUOYI™ can cause fetal harm when administered to a pregnant woman.

Pregnancy Testing

Verify the pregnancy status of females of reproductive potential prior to initiating TUOYI™.

Contraception

Females

Advise females of reproductive potential to use effective contraception during treatment with TUOYI™ and for 4 months after the last dose.

Pediatric Use

The safety and effectiveness of TUOYI™ have not been established in pediatric patients.

Geriatric Use

Of the 146 patients with NPC who were treated with TUOYI™ in combination with cisplatin and gemcitabine 7 (4.8%) were 65 years or older; there were no patients 75 years and older. Clinical studies of TUOYI™ in combination with cisplatin and gemcitabine did not include a

sufficient number of patients aged 65 years and over with NPC to determine whether they respond differently from younger patients.

Of the 851 patients with tumor types including nasopharyngeal carcinoma or other types of tumors from the safety pool treated with TUOYI™ as a single agent, 171 (20%) patients were 65 years or older and 13 (1.5%) patients were 75 years and older. No overall differences in safety were observed between elderly patients and younger patients.

Of the 190 patients with NPC treated with TUOYI™ as single agent, 10 (5%) patients were 65 years or older; there were no patients 75 years and older. Clinical studies of TUOYI™ did not include sufficient numbers of patients aged 65 years and over with NPC to determine whether they respond differently from younger patients.

4.7. Effects on ability to drive and use machines

Toripalimab has minor influence on the ability to drive and use machines. In some patients, dizziness and fatigue have been reported following administration of toripalimab.

4.8. Undesirable effects

The following clinically significant adverse reactions are described elsewhere in the labeling:

- Severe and fatal immune-mediated adverse reactions
- Infusion-related reactions
- Complications of Allogeneic HSCT

Table 3: Adverse reactions in patients treated with toripalimab

Infections and infestations	
Very common	upper respiratory tract infection
Common	pneumonia, urinary tract infection, infection (not specified by site or pathogen), ear infections ¹ , dental and oral soft tissue infections ² , herpes simplex/herpes zoster infection
Uncommon	conjunctivitis, gingivitis, skin and subcutaneous tissue infections ³ , skin infections, bacteraemia, toe infection, paronychia/dermatophytosis of nail, osteomyelitis, pulmonary tuberculosis
Rare	diverticulitis, hepatitis B reactivation, muscle abscess, urosepsis
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	
Common	tumour pain
Uncommon	tumour haemorrhage, tumour rupture
Rare	myelodysplastic syndrome
Blood and lymphatic system disorders	
Very common	anaemia, leukopenia, neutropenia, thrombocytopenia
Common	leukocytosis, neutrophilia, lymphopenia
Uncommon	coagulopathy, bone marrow failure, myelosuppression
Rare	eosinopenia, pancytopenia
Immune system disorders	
Uncommon	hypersensitivity/serum sickness
Endocrine disorders	
Very common	hypothyroidism
Common	hyperthyroidism
Uncommon	thyroiditis, adrenal insufficiency/cortisol decreased, thyroid disorder (excluding hypothyroidism and hyperthyroidism), hypophysitis/empty sella syndrome
Rare	hyperparathyroidism, hypopituitarism

Metabolism and nutrition disorders	
Very common	decreased appetite, hyponatraemia, weight decreased, hypoproteinaemia, hyperglycaemia, hypokalaemia, hyperuricaemia/gout
Common	hypochloraemia, hypomagnesaemia, hypocalcaemia, hypophosphataemia, hyperkalaemia, hypercalcaemia, hypoglycaemia, dehydration
Uncommon	electrolyte imbalance, hyperphosphataemia, hypernatraemia, acid base disorder ⁴ , diabetes mellitus, malnutrition, hypovolaemia
Rare	hypolipidaemia
Psychiatric disorders	
Common	hypersomnia/insomnia
Uncommon	depression/dysphoria, anxiety
Rare	mental disorder, tic
Nervous system disorders	
Very common	neuropathy ⁵
Common	dizziness, headache, neurotoxicity, dysgeusia
Uncommon	somnolence, syncope, encephalopathy, epilepsy, tremor, memory impairment, dysarthria, nervous system disorder, speech disorder
Rare	disturbance in attention, haemorrhage intracranial, paraplegia
Eye disorders	
Common	vision blurred
Uncommon	eye inflammation ⁶ , eye movement disorder, papilloedema
Rare	blepharochalasis, glaucomatocyclitic crises, hypermetropia, retinal haemorrhage
Ear and labyrinth disorders	
Common	ear disorder ⁷
Uncommon	vertigo, deafness
Cardiac disorders	
Very common	arrhythmia ⁸
Uncommon	pericardial effusion, cardiac failure/cardiac dysfunction, myocarditis/immune-mediated myocarditis, myocardial injury/myocardial ischaemia, cardiac discomfort
Rare	aortic valve disease, cardiac disorder
Vascular disorders	
Common	hypertension, hypotension/orthostatic hypotension, embolism and thrombosis
Uncommon	phlebitis
Rare	aortic aneurysm, flushing
Respiratory, thoracic and mediastinal disorders	
Very common	cough
Common	dyspnoea, pneumonitis/immune-mediated lung disease/interstitial lung disease, upper respiratory tract disorders ⁹ , haemoptysis, epistaxis, pleural effusion, hiccups, dysphonia, rhinitis allergic
Uncommon	nasal congestion, respiratory failure, bronchospasm, sinus disorder, pneumonia aspiration, sputum increased, tracheo-oesophageal fistula
Rare	hydrothorax, pleurisy, vocal cord thickening
Gastrointestinal disorders	
Very common	nausea/dyspepsia/eructation, vomiting, constipation/dyschezia, colitis/diarrhoea, abdominal pain
Common	stomatitis, abdominal distension/flatulence, dry mouth, dysphagia, toothache, gastrointestinal haemorrhage, gastrooesophageal reflux disease/hyperchlorhydria
Uncommon	intestinal obstruction/subileus, gastritis, gastroenteritis, oesophageal obstruction, pancreatitis, proctalgia, gastric disorder, gastric ulcer, gastrointestinal disorder, gastric dilatation, gastric fistula, hypoaesthesia oral
Rare	faecaloma, oesophageal ulcer, pancreatic disorder, pneumatosis intestinalis, swollen tongue, tongue discolouration
Hepatobiliary disorders	
Very common	Hyperbilirubinaemia/jaundice

Common	hepatitis ¹⁰ , total bile acids increased
Uncommon	hepatic pain, cholecystitis, hepatic steatosis
Skin and subcutaneous tissue disorders	
Very common	rash ¹¹ , pruritus
Common	alopecia, vitiligo, pigmentation disorder
Uncommon	night sweats, skin disorder, skin exfoliation, hyperhidrosis, dry skin, skin ulcer, hair colour changes, psoriasis, photosensitivity reaction, skin hyperpigmentation
Rare	dermatomyositis, leukoderma, neurodermatitis, onychomadesis, pain of skin, panniculitis, pemphigus, purpura senile, telangiectasia
Musculoskeletal and connective tissue disorders	
Very common	musculoskeletal pain
Common	muscular weakness, arthritis/joint range of motion decreased/peri-arthritis
Uncommon	muscle spasms, intervertebral disc protrusion, myositis
Rare	limb mass
Renal and urinary disorders	
Very common	proteinuria, haematuria
Common	renal injury/nephropathy
Uncommon	pollakiuria, hydronephrosis, pyelocaliectasis, ureteric dilatation
Rare	cystitis noninfective, hydroureter, immune-mediated renal disorder
Reproductive system and breast disorders	
Uncommon	benign prostatic hyperplasia, breast pain, oedema genital, scrotal oedema
Rare	hypomenorrhoea, menorrhagia, menstrual disorder, menstruation irregular, prostatic calcification, vulvovaginal inflammation
General disorders and administration site conditions	
Very common	fatigue, pyrexia, pain ¹²
Common	oedema, influenza like illness, face oedema, chills, eye disorder ¹³
Uncommon	facial pain, swelling, temperature intolerance, thirst
Rare	administration site reactions, hyperplasia, medical device pain, secretion discharge
Investigations	
Very common	liver function test abnormal, thyroid function test abnormal, increased or decreased lipids, urine analysis abnormal ¹⁴
Common	creatinine renal clearance decreased, blood creatine phosphokinase decreased/blood creatine phosphokinase increased, blood lactate dehydrogenase increased, amylase increased, lymphocyte count abnormal/monocyte count abnormal, blood alkaline phosphatase increased, blood urea increased, weight increased, lipase increased, electrocardiogram abnormal, C-reactive protein increased, occult blood positive, cardiac investigation abnormal ¹⁵
Uncommon	platelet count increased, anti-thyroid antibody positive, eosinophil count abnormal, blood prolactin increased, blood testosterone decreased, blood follicle stimulating hormone increased, blood luteinising hormone increased, urine output decreased
Injury, poisoning and procedural complications	
Uncommon	infusion related reaction, contusion/muscle injury, rib fracture

The following terms represent a group of related events that describe a medical condition rather than a single event.

¹ Ear infections include mastoiditis, myringitis, and otitis media.

² Dental and oral soft tissue infections includes oral candidiasis, pericoronitis, and periodontitis.

³ Skin and subcutaneous tissue infections includes cellulitis, folliculitis, and subcutaneous abscess.

⁴ Acid base disorder includes metabolic acidosis, metabolic alkalosis, and metabolic disorder.

⁵ Neuropathy includes anaesthesia, anosmia, formication, hypoaesthesia, Lhermitte's sign, nerve injury, neuropathy peripheral, paraesthesia, parosmia, peripheral motor neuropathy, peripheral sensory neuropathy, peroneal nerve palsy, tongue paralysis, VIth nerve disorder, VIth nerve injury, and vocal cord paralysis.

⁶ Eye inflammation includes eye inflammation, iritis, keratitis, and uveitis.

⁷ Ear disorder includes ear pain, eustachian tube disorder, hypoacusis, middle ear inflammation, otorrhea, and tinnitus.

⁸ Based on a standard query including bradyarrhythmias and tachyarrhythmias.

⁹ Upper respiratory tract disorders includes catarrh, dry throat, laryngeal oedema, laryngeal pain, nasal obstruction, rhinalgia, rhinorrhoea, and throat irritation.

¹⁰ Hepatitis includes drug-induced liver injury, hepatic failure, hepatic function abnormal, and immune-mediated hepatitis.

¹¹ Rash includes dermatitis, dermatitis acneiform, dermatitis allergic, drug eruption, eczema, erythema, erythema multiforme, hand dermatitis, palmar-plantar erythrodysesthesia syndrome, papule, rash, rash erythematous, rash generalised, rash maculo-papular, rash papular, rash pruritic, rash pustular, rash vesicular, skin plaque, and urticaria.

¹² Pain includes chest discomfort, chest pain, eye pain, lymph node pain, non-cardiac chest pain, and pain.

¹³ Eye disorder includes cataract, diplopia, dry eye, eye pruritus, and eye swelling.

¹⁴ Urine analysis abnormal includes bilirubin urine present, crystal urine present, glucose urine present, urea urine increased, urinary casts, urinary casts present, urinary sediment present, urine bilirubin increased, urine ketone body present, urobilinogen urine increased, and white blood cells urine positive.

¹⁵ Cardiac investigations abnormal includes blood creatine phosphokinase MB increased, brain natriuretic peptide increased, and troponin increased.

Clinical Trials Experience

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in practice.

The data described in the WARNINGS AND PRECAUTIONS section reflect exposure to TUOYI™ at a dose of 240 mg every 3 weeks in combination with up to 6 cycles of cisplatin and gemcitabine, followed by TUOYI™ 240 mg IV every 3 weeks, in 146 patients with NPC enrolled in a randomized, double-blind, placebo-controlled trial (JUPITER-02). Among the 146 patients, 73% were exposed to TUOYI™ for 6 months or more and 54% were exposed for 12 months or more. The most common adverse reactions ($\geq 20\%$) were: nausea (71%), vomiting (68%), decreased appetite (55%), constipation (39%), hypothyroidism (38%), rash (36%), pyrexia (32%), diarrhea (31%), peripheral neuropathy (30%), cough (26%), musculoskeletal pain (25%), upper respiratory infection (23%), insomnia (23%), dizziness (21%), and malaise (21%). The most common Grade 3 or 4 laboratory abnormalities ($\geq 2\%$) were: decreased neutrophils (58%), decreased lymphocytes (57%), decreased hemoglobin (50%), decreased platelets (33%), decreased potassium (10%), decreased sodium (9%), increased alanine aminotransferase (6%), increased or decreased magnesium (4.2% each), decreased calcium (3.5%), increased aspartate aminotransferase (2.7%), and bilirubin increased (2.1%).

The data described in the WARNINGS AND PRECAUTIONS section also reflects exposure to TUOYI™ as a single agent at a dose of 3 mg/kg IV every 2 weeks in 851 patients enrolled in 12 trials: one randomized, active-controlled trial and 11 open-label, non-randomized trials. The tumor types included nasopharyngeal carcinoma (n=193) or other types of tumors (n=658). Among the 851 patients treated with TUOYI™ as a single agent, 35% were exposed for 6 months or more and 20% were exposed for 12 months or more. In this pooled safety population, the most common ($\geq 20\%$) adverse reactions were: fatigue (22%), hypothyroidism (20%), and musculoskeletal pain (20%). The most common Grade 3 or 4 laboratory abnormalities ($\geq 2\%$), were decreased sodium (9%), decreased lymphocytes (8%), decreased hemoglobin (7%), decreased fibrinogen (4.5%), increased lipase (4.0%), increased amylase (2.9%), decreased

phosphate (2.8%), increased aspartate aminotransferase (2.6%), increased glucose (2.5%), and increased triglycerides (2.1%).

First-line Treatment of Metastatic or Recurrent, Locally Advanced Nasopharyngeal Carcinoma (NPC)

The safety of TUOYI™ in combination with cisplatin and gemcitabine was evaluated in JUPITER-02 [see Clinical Studies (14)]. Key eligibility criteria were recurrent locally advanced or metastatic nasopharyngeal carcinoma (NPC) not previously treated with systemic chemotherapy for recurrent or metastatic disease. Patients with recurrent NPC after treatment with curative intent were required to have an interval of at least 6 months between the last dose of radiotherapy or chemotherapy and recurrence. Patients received TUOYI™ 240 mg (n=146) or placebo intravenously (IV) every 3 weeks (n=143), in combination with cisplatin 80 mg/m² IV every 3 weeks and gemcitabine 1000 mg/m² IV days 1 and 8 for up to 6 cycles followed by TUOYI™ 240 mg or placebo IV every 3 weeks until disease progression, unacceptable toxicity, or completion of 2 years of treatment. Among patients who received TUOYI™, 73% were exposed for 6 months or longer and 54% were exposed for greater than one year.

The median age of patients who received TUOYI™ was 48 years (range: 19 to 72), 83% male, 100% Asian, 60% had recurrent disease, and 40% presented with metastatic disease. Eastern Cooperative Oncology Group (ECOG) Performance Status (PS) was 0 (57%) or 1 (43%). Approximately 59% of patients had received at least one prior systemic therapy for locally advanced disease and 60% had received prior radiation therapy.

Serious adverse reactions occurred in 43% of patients receiving TUOYI™ in combination with cisplatin and gemcitabine. Serious adverse drug reactions in $\geq 2\%$ were thrombocytopenia (14%), neutrophil count decreased (10%), pneumonia (10%), anemia (9%), abnormal hepatic function (2.7%), and rash (2.1%). Of the patients who received TUOYI™ in combination with cisplatin and gemcitabine, there were three fatal adverse reactions (2.1%) one due to epistaxis; one due to intracranial hemorrhage associated with immune-related thrombocytopenia and coagulopathy; and one due to pneumonia.

Permanent discontinuation of TUOYI™, when given in combination with cisplatin and gemcitabine, due to an adverse reaction occurred in 12% of patients. Adverse reactions resulting in permanent discontinuation of TUOYI™ in $\geq 1\%$ were pneumonia (2.1%), pulmonary tuberculosis (1.4%), rash (1.4%), and vomiting (1.4%).

Dosage interruptions of TUOYI™ due to an adverse reaction occurred in 50% of patients. Adverse reactions which required dosage interruption in $\geq 2\%$ were anemia (17%), decreased neutrophils (12%), thrombocytopenia (12%), acute kidney injury (4.1%), pneumonia (6%), fatigue (2.7%), upper respiratory infection (2.7%), and hypothyroidism (2.1%).

Table 4 summarizes the adverse reactions in JUPITER-02.

Table 4: Adverse Reactions ($\geq 10\%$) in Patients with Recurrent, Locally Advanced or Metastatic NPC Who Received TUOYI™ in Combination with Cisplatin and Gemcitabine in JUPITER-02

Adverse Reaction ¹	TUOYI™ Cisplatin/Gemcitabine N = 146		Placebo Cisplatin/Gemcitabine N = 143	
	All Grades (%)	Grade 3 or 4 (%)	All Grades (%)	Grade 3 or 4 (%)
Gastrointestinal Disorders				

Adverse Reaction ¹	TUOYI™ Cisplatin/Gemcitabine N = 146		Placebo Cisplatin/Gemcitabine N = 143	
	All Grades (%)	Grade 3 or 4 (%)	All Grades (%)	Grade 3 or 4 (%)
Nausea	71	1.4	84	2.8
Vomiting	68	2.1	66	2.1
Constipation	39	0	46	0
Diarrhea	31	1.4	23	0
Stomatitis ²	12	0	8	0.7
Metabolism and Nutrition Disorders				
Decreased appetite	55	0.7	63	0
Endocrine Disorders				
Hypothyroidism ³	38	0.7	17	0
Skin Disorders				
Rash ⁴	36	3.4	28	2.8
Pruritus	17	0	8	0
General Disorders				
Pyrexia	32	1.4	24	0.7
Malaise	21	0.7	20	0
Fatigue ⁵	19	0.7	22	2.1
Nervous System Disorders				
Peripheral neuropathy ⁶	30	0	31	0.7
Dizziness	21	0	22	0.7
Headache	18	0	23	0.7
Respiratory, Thoracic and Mediastinal Disorders				
Cough ⁷	26	0	27	0
Musculoskeletal Disorders				
Musculoskeletal pain ⁸	25	0	25	0.7
Infections				
Upper respiratory infection ⁹	23	3.4	13	2.8
Pneumonia ¹⁰	18	11	7	3.5
Psychiatric Disorders				
Insomnia	23	0	17	0
Vascular Disorders				
Epistaxis	10	1.3	13	2.8
Hypertension ¹¹	10	6	6	4.2

¹ NCI CTCAE v5.0.

² Includes mouth ulceration, stomatitis, and radiation stomatitis.

³ Includes hypothyroidism, tri-iodothyronine decreased, tri-iodothyronine free decreased, and thyroiditis.

⁴ Includes acneiform dermatitis, allergic dermatitis, catheter-site rash, dermatitis, drug eruption, eczema, erythema, macule, maculopapular rash, palmar-plantar erythrodysesthesia syndrome, papule, pruritic rash, rash, and urticaria.

⁵ Includes asthenia and fatigue.

⁶ Includes hypoesthesia, neuralgia, neuropathy peripheral, paresthesia, peripheral sensory neuropathy.

⁷ Includes cough and productive cough.

⁸ Includes back pain, bone pain, musculoskeletal chest pain, musculoskeletal pain, myalgia, neck pain, pain in extremity, pain in jaw.

⁹ Includes acute sinusitis, bronchitis, laryngitis, nasopharyngitis, pharyngitis, respiratory tract infection, rhinitis, sinusitis, and upper respiratory tract infection.

¹⁰ Includes aspiration pneumonia and pneumonia

¹¹ Includes blood pressure increased, blood pressure systolic increased, hypertension, and hypertensive crisis.

Table 5 summarizes the laboratory abnormalities in JUPITER-02.

Table 5: Select Laboratory Abnormalities ($\geq 20\%$) That Worsened from Baseline in Patients with Recurrent, Locally Advanced or Metastatic NPC Who Received TUOYI™ in Combination with Cisplatin and Gemcitabine in JUPITER-02

Laboratory Abnormalities*	TUOYI™ Cisplatin/Gemcitabine		Placebo Cisplatin/Gemcitabine	
	All Grades† (%)	Grade 3 or 4 (%)	All Grades (%)	Grades 3 or 4 (%)
Hematology				
Decreased hemoglobin	94	50	97	38
Decreased neutrophils	91	58	95	63
Decreased lymphocytes	88	57	88	49
Decreased platelets	71	33	66	31
Chemistry				
Decreased magnesium	78	4.2	77	8
Decreased sodium	63	9	62	6
Increased alanine aminotransferase	58	6	50	3.5
Increased aspartate aminotransferase	58	2.7	53	4.9
Decreased albumin	49	0	48	0
Decreased calcium	45	3.5	46	4.2
Increased lactate dehydrogenase	42	0	35	0
Increased calcium	39	0	35	0.7
Decreased potassium	40	10	39	8
Increased creatinine	39	0.7	41	0
Increased alkaline phosphatase	27	0	27	0
Decreased glucose	23	1.4	16	0

* Each test incidence is based on the number of patients who had both baseline and at least one on-study laboratory measurement available: TUOYI™ /chemotherapy (range: 139 to 146 patients) and placebo/chemotherapy (range: 136 to 143 patients). † Graded per NCI CTCAE v5.0; AKP=alkaline phosphatase. ALT=alanine aminotransferase. AST=aspartate aminotransferase.

Previously Treated, Unresectable or Metastatic Nasopharyngeal Carcinoma (NPC)

The safety of TUOYI™ was evaluated in POLARIS-02. Eligible patients had previously treated unresectable or metastatic NPC. Patients received TUOYI™ 3 mg/kg every 2 weeks as an intravenous infusion until disease progression or unacceptable toxicity. Among patients who received TUOYI™, 33% were exposed for 6 months or longer and 21% were exposed for greater than one year.

The median age of patients who received TUOYI™ was 46 years (range: 22 to 71), 83% male, 100% Asian, Eastern Cooperative Oncology Group (ECOG) Performance Status (PS) of 0 (35%) or 1 (65%) and median weight 59 kg (range: 32 to 101 kg).

Serious adverse reactions occurred in 24% of patients who received TUOYI™. Serious adverse drug reactions in ($\geq 2\%$) were pneumonia (4.7%), abnormal hepatic function (2.6%), and

hyperbilirubinemia (2.1%). Fatal adverse reactions occurred in 3.7% of patients who received TUOYI™, including death not otherwise specified (1.6%), tumor hemorrhage (0.5%), hepatic failure and thrombocytopenia (0.5%), hyponatremia (0.5%), and sudden death (0.5%).

Permanent discontinuation of TUOYI™ due to an adverse reaction occurred in 9% of patients. Adverse reaction resulting in permanent discontinuation of TUOYI™ in ≥1% included pneumonia (1.1%), abnormal hepatic function (1.1%), and hyperbilirubinemia (1.1%).

Dosage interruptions due to an adverse reaction occurred in 23% of patients. Adverse reactions which required dosage interruption in ≥1% were pneumonia (2.1%), thrombocytopenia (2.1%), fatigue (1.6%), hyperbilirubinemia (1.6%), anemia (1.1%), decreased appetite (1.1%), abnormal hepatic function (1.1%), hypothyroidism (1.1%), and pneumonitis (1.1%).

Table 6 summarizes the adverse reactions in POLARIS-02.

Table 6: Adverse Reactions (≥10%) in Patients with Previously Treated, Unresectable or Metastatic NPC Who Received TUOYI™ in POLARIS-02

Adverse Reaction*	TUOYI™ N=190	
	All Grades (%)	Grade 3 or 4 (%)
Endocrine Disorders		
Hypothyroidism ¹	27	0
General Disorders		
Fatigue ²	22	2.6
Pyrexia	16	0
Respiratory Disorders		
Cough ³	20	0
Musculoskeletal Disorders		
Musculoskeletal Pain ⁴	18	1.1
Metabolism and Nutrition		
Decreased Appetite	13	1.1
Gastrointestinal Disorders		
Constipation	11	0
Skin and Subcutaneous Disorders		
Pruritus	11	0
Rash ⁵	11	0
Investigations		
Weight Decreased	11	0

* Toxicity was graded per National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) v4.03.

¹ Includes hypothyroidism, thyroiditis, tri-iodothyronine decreased, and tri-iodothyronine free decreased

² Includes fatigue and asthenia

³ Includes cough and productive cough.

⁴ Includes musculoskeletal pain and myalgia.

⁵ Includes dermatitis allergic, eczema, and rash.

Table 7 summarizes the laboratory abnormalities in POLARIS-02.

Table 7: Select Laboratory Abnormalities ($\geq 20\%$) That Worsened from Baseline in Patients with Previously Treated, Unresectable or Metastatic NPC Who Received TUOYI™ in POLARIS-02

	TUOYI™	
	All Grades (%) ¹	Grade 3 or 4 (%) ¹
Chemistry		
Decreased albumin	38	0.5
Decreased sodium	35	11
Decreased phosphate	32	3.2
Increased aspartate aminotransferase	30	3.8
Decreased calcium	29	0.5
Increased alkaline phosphatase	28	2.2
Increased triglyceride	26	1.1
Increased glucose	24	1.1
Increased alanine aminotransferase	23	1.6
Hematology		
Decreased lymphocytes	52	9
Decreased hemoglobin	43	6

¹ Toxicity graded per NCI CTCAE v4.03. The denominator used to calculate the rate varied from 141 to 186 based on the number of patients with a baseline value and at least one post-treatment value.

4.9. Overdose

In case of overdose, patients should be closely monitored for signs or symptoms of adverse reactions and appropriate symptomatic treatment instituted.

5. PHARMACOLOGICAL PROPERTIES

5.1. Pharmacodynamic properties

Pharmacotherapeutic group: Antineoplastic agents, ATC code: L01FF13

Mechanism of Action

Binding of the PD-1 ligands, PD-L1 and PD-L2, to the PD-1 receptor found on T cells, inhibits T cell proliferation and cytokine production. Upregulation of PD-1 ligands occurs in some tumors and signaling through this pathway can contribute to inhibition of active T-cell immune surveillance of tumors. Toripalimab is a humanized IgG4 monoclonal antibody that binds to the PD-1 receptor and blocks its interaction with PD-L1 and PD-L2, releasing PD-1 pathway-mediated inhibition of the immune response, including the anti-tumor immune response. In syngeneic mouse tumor models, blocking PD-1 activity resulted in decreased tumor growth.

Pharmacodynamics

The toripalimab exposure-response relationships and time course of pharmacodynamic response are not fully characterized.

Clinical Studies

First-line Treatment of Metastatic or Recurrent, Locally Advanced NPC with Cisplatin and Gemcitabine

The efficacy of TUOYI™ in combination with cisplatin and gemcitabine was investigated in JUPITER-02 (NCT03581786), a randomized, multicentre, single region, double-blind, placebo-controlled trial in 289 patients with metastatic or recurrent, locally advanced NPC who had not received previous systemic chemotherapy for recurrent or metastatic disease. Patients with recurrent NPC after treatment with curative intent were required to have an interval of at least 6 months between the last dose of radiotherapy or chemotherapy and recurrence. Patients with autoimmune disease, other than stable hypothyroidism or Type I diabetes, and patients who required systemic immunosuppression were ineligible.

Randomization was stratified according to Eastern Cooperative Oncology Group (ECOG) Performance Status (PS) (0 versus 1) and disease stage (recurrent versus metastatic). Patients were randomized (1:1) to receive one of the following treatments:

- TUOYI™ 240 mg intravenously every 3 weeks in combination with cisplatin 80 mg/m² on Day 1 every 3 weeks gemcitabine 1000 mg/m² on Days 1 and 8 for up to 6 cycles, followed by TUOYI™ 240 mg once every 3 weeks, or
- Placebo intravenously every 3 weeks in combination with cisplatin 80 mg/m² on Day 1 every 3 weeks and gemcitabine 1000 mg/m² on Days 1 and 8 for up to 6 cycles, followed by placebo once every 3 weeks.

Treatment with TUOYI™ or placebo continued until disease progression per RECIST v1.1, unacceptable toxicity, or a maximum of 2 years. Administration of TUOYI™ was permitted beyond radiographic progression if the patient was deriving benefit as assessed by the investigator. Tumor assessments were performed every 6 weeks for the first 12 months and every 9 weeks thereafter. The main efficacy outcome measure was Blinded Independent Review Committee (BIRC)-assessed progression-free survival (PFS) according to RECIST v1.1. Additional efficacy outcome measures include BIRC-assessed overall response rate (ORR) and overall survival (OS).

The study population characteristics were: median age of 48 years (range: 19 to 72), 4.8% age 65 or older, 83% male, 100% Asian, and 57% had ECOG PS of 0. Eighty-six percent of patients had metastatic disease at study entry. Histological subtypes of NPC included 98% non-keratinizing, 1% keratinizing squamous cell carcinoma, and 1% did not have the subtype identified.

Efficacy results of the pre-specified interim analysis of PFS and final analysis of OS are summarized in Table 8 and Figure 1 below. The trial demonstrated statistically significant improvements in BIRC-assessed PFS, ORR and OS for patients randomized to TUOYI™ in combination with cisplatin/gemcitabine compared to cisplatin and gemcitabine with placebo.

Table 8: Efficacy Results in JUPITER-02

Endpoints	TUOYI™ Cisplatin/Gemcitabine N=146	Placebo Cisplatin/Gemcitabine N=143
BIRC-Assessed Progression-free Survival¹		
Number of Events, n (%)	49 (34)	79 (55)
Median, months (95% CI)	11.7 (11.0, NE)	8.0 (7.0, 9.5)
Hazard Ratio (95% CI) ³	0.52 (0.36, 0.74)	
p-value ⁴	0.0003	

Endpoints	TUOYI™ Cisplatin/Gemcitabine N=146	Placebo Cisplatin/Gemcitabine N=143
BIRC-Assessed Overall Response Rate ¹		
ORR, % (95% CI)	77 (70, 84)	66 (58, 74)
Complete Response Rate (%)	19	11
Partial Response Rate (%)	58	55
p-value ⁵	0.0353	
BIRC-Assessed Duration of Response		
Median, months (95% CI)	10.0 (8.8, NE)	5.7 (5.4, 6.8)
Overall Survival ²		
Number of Deaths, n (%)	57 (39)	76 (53)
Median, months (95% CI)	NR (38.7, NE)	33.7 (27.0, 44.2)
Hazard Ratio (95% CI) ³	0.63 (0.45, 0.89)	
p-value ⁶	0.0083	

¹ PFS and ORR results are based on the pre-specified interim analysis with data cutoff of May 30, 2020.

² OS results are based on the final analysis with a data cutoff of November 18, 2022.

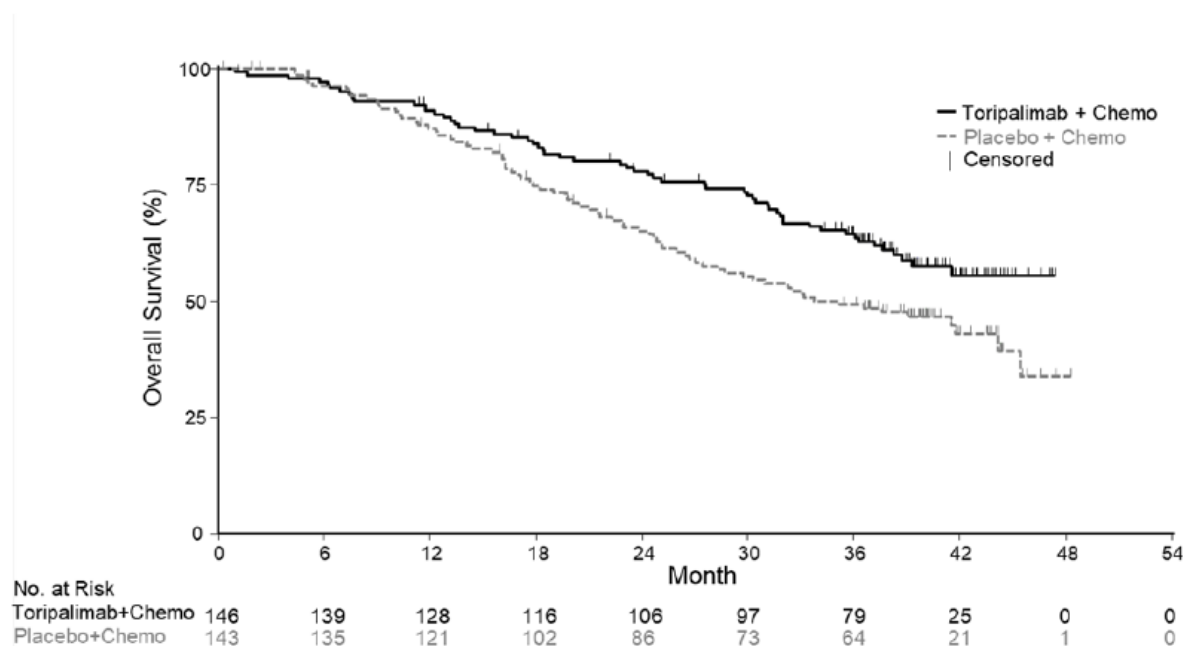
³ Based on the stratified Cox proportional-hazards model using the stratification factors at randomization, ECOG performance status and disease stage.

⁴ Two-sided p-value, based on the stratified log-rank test, as compared with an alpha boundary of 0.010.

⁵ Two-sided p-value, based upon Cochran-Mantel-Haenszel test.

⁶ Two-sided p-value, based on the stratified log-rank test, as compared with an alpha boundary of 0.049995. BIRC=blinded independent review committee; CI= confidence interval; NR=Not Reached; NE=Not estimable.

Figure 1: Kaplan-Meier Curves of Overall Survival for JUPITER-02



Previously Treated Unresectable or Metastatic NPC

The efficacy of TUOYI™ was investigated in POLARIS-02 (NCT 02915432), an open-label, multicentre, multicohort trial conducted in a single country. The trial included a total of 172 patients with unresectable or metastatic NPC who had received prior platinum-based chemotherapy for treatment of recurrent or metastatic NPC or had disease progression within 6 months of completion of platinum-based chemotherapy administered as neoadjuvant, adjuvant, or definitive chemoradiation treatment for locally advanced disease. Key exclusion criteria included previous treatment with an anti-PD-(L)1 antibody; active autoimmune disease or other medical conditions requiring immunosuppressive therapy. Patients received TUOYI™ 3 mg/kg intravenously every 2 weeks until disease progression per RECIST v1.1 or unacceptable toxicity. Tumor response assessments were performed every 8 weeks for the first year and every 12 weeks thereafter. The major efficacy outcome measures were confirmed ORR and duration of response (DOR) as assessed by a Blinded Independent Review Committee (BIRC) using RECIST v1.1.

The median age was 45 years (range: 22 to 68), 4.1% age 65 or older, 83% male, 100% Asian, and Eastern Cooperative Oncology Group (ECOG) Performance Status (PS) of 0 (37%). Patients had received a median of 2 prior systemic therapies for recurrent/metastatic disease (range: 1-13). Ninety-five percent of patients had metastatic disease, 95% had non-keratinizing NPC, 2.9% had keratinizing squamous cell carcinoma and 1.7% did not have the subtype identified.

Efficacy results for POLARIS-02 are summarized in Table 9 below.

Table 9: Efficacy Results for POLARIS-02

Endpoint	TUOYI™ (N=172)
BIRC-Assessed Overall Response Rate¹	
Overall Response Rate, % (95% CI)	21 (15, 28)
Complete Response Rate, %	2.3
Partial Response Rate, %	19
BIRC-Assessed Duration of Response (DOR)	
Median, months (95% CI)	14.9 (10.3, NE)
Patients with DOR > 6 months ² , n (%)	30 (83%)
Patients with DOR > 12 months ² , n (%)	14 (39%)

CI=confidence interval. n=number. NE = not estimable.

¹ Confirmed overall response rate assessed by BIRC

² Based on observed duration of response

BIRC=blinded independent review committee

5.2. Pharmacokinetic properties

Toripalimab pharmacokinetic parameters are presented as geometric mean (coefficient of variation [CV]%) unless otherwise noted. Toripalimab concentrations increased in nonlinearly over the dose range of 0.3 to 10 mg/kg every two weeks (0.1 to 3.3 times the approved recommended 3 mg/kg dosage in a 64 kg patient). Steady state was reached by Week 7. The mean accumulation ratio was approximately 1.4 for maximum concentration (C_{max}) and 1.9 for area under the serum concentration curve (AUC) following multiple doses at the approved recommended dosages of 240 mg Q3W in combination with cisplatin and gemcitabine and 3 mg/kg Q2W as monotherapy.

Distribution

The mean volume of distribution at steady state (V_{ss}) of toripalimab was 3.7 L (27%).

Elimination

The mean clearance (CL) was 14.9 mL/h (31%) after the first dose and 9.5 mL/h (36%) at steady state. The mean terminal half-life ($t_{1/2}$) ($\pm$ standard deviation) was 10 ± 1.5 days after the first dose and 18 ± 9.4 days at steady state.

Metabolism

Toripalimab is expected to be metabolized into small peptides by catabolic pathways.

Specific Populations

No clinically significant differences in the pharmacokinetics were observed based on age (21 to 85 years), body weight (32 to 164 kg), sex, race (White and Asian), concomitant chemotherapy, mild renal impairment (creatinine clearance [CL_{Cr}] 60 to 89 mL/min), mild hepatic impairment (total bilirubin > 1 to 1.5 times ULN with any AST or total bilirubin $\leq$ ULN with AST $>$ ULN), tumor burden and primary cancer.

The effect of moderate (total bilirubin > 1.5 to 3 times ULN and any AST) or severe (total bilirubin > 3 times ULN and any AST) hepatic impairment or of moderate (CL_{Cr} 30 to 59 mL/min) or severe (CL_{Cr} 15 to 29 mL/min) renal impairment on the pharmacokinetics of toripalimab has not been studied.

5.3. Immunogenicity

The observed incidence of anti-drug antibodies (ADA) is highly dependent on the sensitivity and specificity of the assay. Differences in assay methods preclude meaningful comparisons of the incidence of ADA in the studies described below with the incidence of ADA in other studies, including those of TUOYI™ or of other toripalimab products.

Of the 146 evaluable patients in JUPITER-02 with nasopharyngeal cancer who received TUOYI™ 240 mg every 3 weeks for a median duration of 15.1 months, in combination with gemcitabine and cisplatin, 3.4% tested positive for treatment-emergent ADA. Of the 190 evaluable patients in study POLARIS-02 with nasopharyngeal cancer who received TUOYI™ 3 mg/kg every 2 weeks for a median duration of 3.3 months, 3.7% of patients developed treatment-emergent ADA. Neutralizing antibodies have not been tested.

Due to the low incidence of ADA, the effect of these antibodies on the pharmacokinetics, pharmacodynamics, safety, or effectiveness of TUOYI™ is unknown.

5.4. Preclinical safety data

Carcinogenesis, Mutagenesis, Impairment of Fertility

No studies have been performed to test the potential of toripalimab for carcinogenicity or genotoxicity.

Fertility studies have not been conducted with toripalimab. In 4-week and 26-week repeat-dose toxicology studies in sexually mature cynomolgus monkeys, there were no adverse or notable effects in the male and female reproductive organs.

Animal Toxicology and/or Pharmacology

In animal models, inhibition of PD-1/PD-L1 signaling increased the severity of some infections and enhanced inflammatory responses. Mycobacterium tuberculosis-infected PD-1 knockout mice exhibit markedly decreased survival compared with wild-type controls, which correlated with increased bacterial proliferation and inflammatory responses in these animals. PD-1 blockade using a primate anti-PD1 antibody was also shown to exacerbate M. tuberculosis infection in rhesus macaques. PD-1 and PD-L1 knockout mice and mice receiving PD-L1 blocking antibody have also shown decreased survival following infection with lymphocytic choriomeningitis virus.

6. PHARMACEUTICAL PARTICULARS

6.1. List of excipients

Citric acid monohydrate
Mannitol
Polysorbate 80
Sodium chloride
Sodium citrate
Water for Injection.

6.2. Incompatibilities

This medicinal product must not be mixed with other medicinal products except those mentioned in section 6.6.

6.3. Shelf life

3 years.

6.4. Special precautions for storage

TUOYI™ (toripalimab), injection is a clear to slightly opalescent, colourless to slightly yellow solution supplied in a carton containing one 240 mg/6 mL (40 mg/mL) single-dose vial. Store vials refrigerated at 2°C to 8°C in original carton to protect from light. Do not freeze. Do not shake.

6.5. Nature and contents of container

Type 1 neutral borosilicate glass vial capped sealed with a chlorobutyl rubber stopper and sealed with a 20 mm flip-off seal (aluminium), containing 6 mL of concentrate for solution for infusion.
Each carton contains one vial.

7. MANUFACTURER

Suzhou Union Biopharm Co., Ltd.
No. 999, Longqiao Road,
Wujiang Economic and Technological Development Zone,
Suzhou City, Jiangsu Province, China

8. PRODUCT REGISTRATION HOLDER

Rxilient Health Malaysia Sdn. Bhd.
L12-01, KYM Tower, No. 8 Jalan PJU 7/6,
Mutiarra Damansara, 47800 Petaling Jaya,
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9. DATE OF REVISION

06 March 2026